

MIROSLAWIEC, Poland

WMO: 122260

Lat: 53.395N

Lon: 16.083E

Elev: 495

StdP: 14.43

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	11.2	1.0	5.9	8.3	6.1	7.7	12.9	21.9	39.1	19.8	36.5	4.6	120	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	85.3	66.5	81.9	65.1	78.6	63.7	69.2	80.6	67.4	77.7	65.7	74.4	8.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	96.1	72.6	64.0	90.9	71.0	62.4	85.8	69.2	33.6	81.1	32.1	78.0	30.8	74.1	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.6	14.7	DB	-0.4	91.6	7.3	3.5	-5.6	94.2	-9.9	96.2	-14.0	98.2	-19.4	100.7	(4)
(5)			WB	-0.7	72.9	7.4	1.8	-6.1	74.2	-10.4	75.3	-14.5	76.3	-19.9	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	28.3	31.3	37.2	46.8	54.7	60.8	65.1	64.1	56.1	46.9	39.0	32.7	(6)
(7)		DBStd	14.53	9.43	8.52	8.05	7.42	7.44	6.24	5.83	5.68	5.77	6.42	6.10	8.19	(7)
(8)		HDD50	2797	674	524	400	145	35	1	0	0	11	138	332	537	(8)
(9)		HDD65	6760	1139	944	862	547	329	158	73	88	275	562	781	1002	(9)
(10)		CDD50	1695	0	0	3	48	181	325	467	436	193	41	1	0	(10)
(11)		CDD65	183	0	0	0	0	10	31	76	59	7	0	0	0	(11)
(12)		CDH74	1961	0	0	0	16	146	381	805	537	76	0	0	0	(12)
(13)		CDH80	515	0	0	0	2	25	93	239	142	14	0	0	0	(13)
(14)	Wind	WSAvg	6.9	7.4	7.6	7.3	7.3	6.7	6.8	6.4	6.0	6.2	6.6	7.1	7.5	(14)
(15)	Precipitation	PrecAvg	26.6	2.0	1.5	1.9	1.5	2.5	2.8	3.2	3.0	2.4	1.9	2.0	2.0	(15)
(16)		PrecMax	41.5	4.9	3.2	4.0	3.6	5.1	8.6	6.1	7.0	5.9	4.9	4.0	3.8	(16)
(17)		PrecMin	18.9	0.2	0.2	0.4	0.3	1.3	0.9	0.6	0.5	0.6	0.6	0.1	0.4	(17)
(18)		PrecStd	5.2	1.3	0.7	0.9	0.8	1.1	1.5	1.5	1.5	1.0	1.1	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.0	51.6	63.5	75.5	83.4	88.1	90.6	89.9	82.0	67.8	54.9	52.3	(19)
MCWB			46.9	47.9	50.7	57.3	64.2	69.1	67.5	69.0	66.3	58.5	51.4	49.2	(20)	
2%		DB	45.9	48.2	57.6	70.1	78.1	82.2	85.7	83.8	75.3	63.9	51.5	48.2	(21)	
		MCWB	44.0	44.4	46.8	55.1	61.7	64.6	66.5	66.0	61.9	57.1	49.1	46.5	(22)	
5%		DB	43.1	45.2	52.8	65.6	73.8	78.5	82.4	79.7	71.5	60.3	49.4	45.5	(23)	
		MCWB	41.5	42.3	45.6	52.5	59.8	62.9	65.5	64.6	59.8	55.0	47.7	43.6	(24)	
10%		DB	40.4	42.4	48.8	61.3	69.4	74.2	78.4	76.1	67.3	57.1	47.6	42.8	(25)	
		MCWB	39.1	39.9	43.8	50.2	57.4	61.3	64.4	63.5	58.4	53.6	46.1	41.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	48.8	52.1	59.9	66.7	71.6	71.1	71.0	68.4	60.6	52.0	50.0	(27)
MCD B			50.2	50.7	59.7	73.0	77.2	84.2	82.5	83.0	80.2	65.7	53.8	51.8	(28)	
2%		WB	44.2	45.4	49.2	56.5	63.7	67.2	69.1	68.7	63.7	57.9	49.5	46.8	(29)	
		MCD B	45.8	47.2	54.6	66.6	75.0	77.7	80.8	79.1	71.1	61.9	50.8	48.3	(30)	
5%		WB	41.6	42.6	46.6	53.8	61.4	64.9	67.5	66.9	61.6	55.8	47.9	43.8	(31)	
		MCD B	43.0	44.8	51.6	63.2	71.0	74.4	78.6	75.5	68.6	59.6	49.2	45.1	(32)	
10%		WB	39.1	40.0	44.3	51.3	59.1	62.9	65.8	65.3	59.7	53.8	46.3	41.4	(33)	
		MCD B	40.5	42.2	48.3	59.5	67.4	71.3	74.7	72.8	65.8	56.9	47.5	42.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	10.8	14.9	20.2	20.2	19.7	19.1	19.2	17.6	13.8	8.5	7.7	(35)
MCDBR			8.8	13.6	22.9	27.3	26.1	26.3	26.1	25.1	24.2	19.0	10.7	9.5	(36)	
MCWBR			7.8	10.3	14.1	14.9	12.7	12.0	10.1	11.1	12.4	12.4	8.8	8.2	(37)	
5% WB		MCDBR	8.8	12.2	19.2	24.7	23.3	22.5	22.1	20.5	21.0	16.5	9.9	9.2	(38)	
		MCWBR	8.1	10.0	12.9	14.2	12.3	11.9	10.3	10.6	12.2	11.6	8.6	8.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.333	0.366	0.405	0.403	0.399	0.422	0.419	0.377	0.359	0.338	0.303	(40)	
taud		2.299	2.302	2.269	2.226	2.281	2.316	2.266	2.282	2.381	2.402	2.365	2.313	(41)		
Ebn at Noon		209	241	257	261	268	270	261	254	252	229	196	186	(42)		
Edh at Noon		19	26	33	39	39	38	40	37	30	24	18	16	(43)		
(44)	All-Sky Solar Radiation	RadAvg	188	405	782	1311	1627	1708	1587	1367	995	528	214	131	(44)	
(45)		RadStd	25	59	106	180	187	143	206	130	116	78	30	13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-263	+113	N/A

Nomenclature: See separate page